



LECTURE 10

VIRUSES AND VIRAL INFECTIONS

By

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Intended Learning Outcomes (ILOs)

By the end of this lecture, students will be able to:



1. Understand the **basic structure** and **components** of **viruses**.



2. **Classify viruses** according to their **type** and **characteristics**.



3. Explain how common viral diseases occur and spread.



4. Identify key **prevention** and **vaccination** measures.

Introduction to Viruses



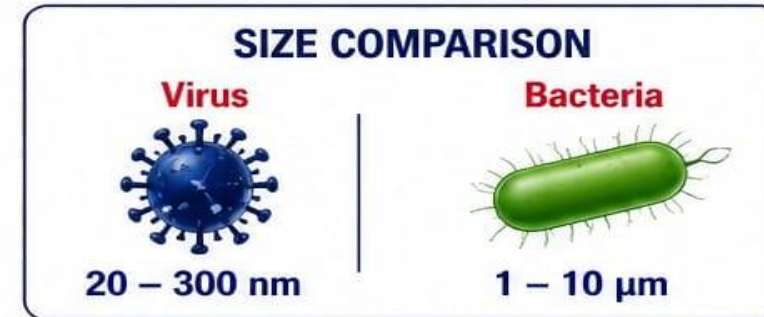
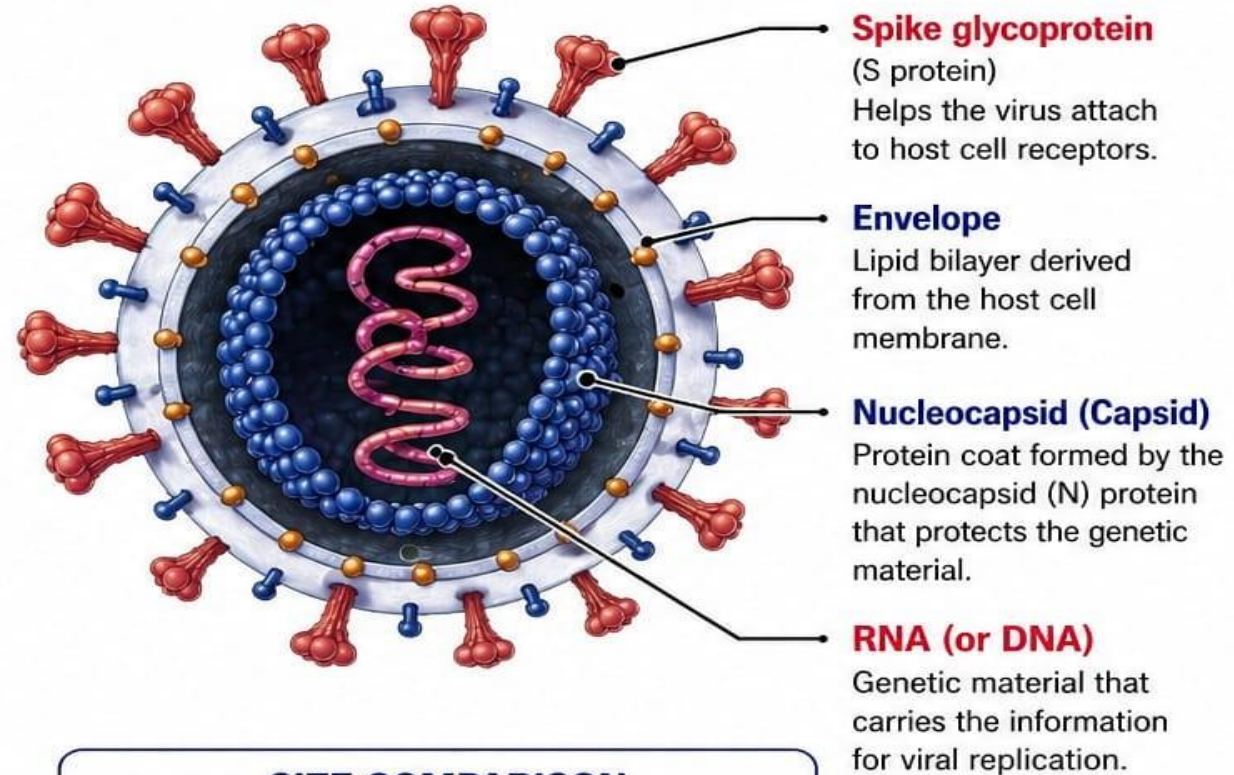
DEFINITION

- Viruses are **obligate intracellular parasites**
- Cannot live or reproduce **outside a host cell**
- Much **smaller** than bacteria



KEY CHARACTERISTICS

- **Non-cellular** (no cytoplasm, no organelles)
- Contain **DNA or RNA** (not both)
- Require **host machinery** for replication

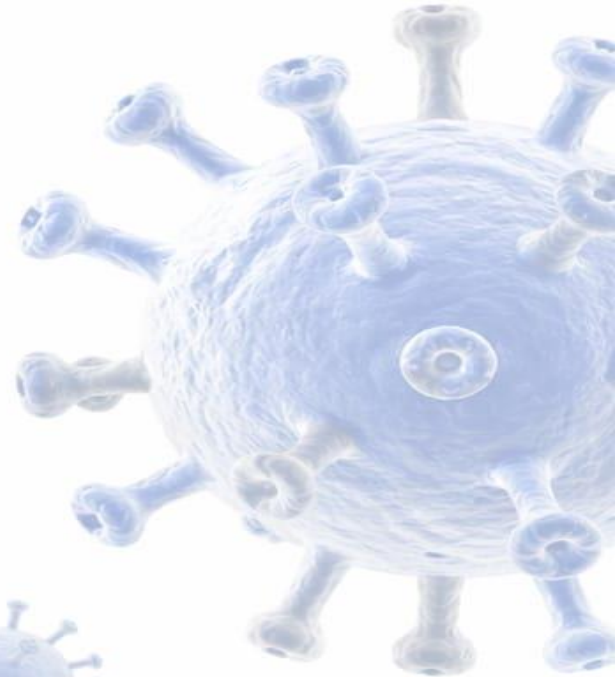


Structure of viruses



1- Viral nucleic acid (genome):

- Viruses have **one type only** of nucleic acid, either **DNA** or **RNA**. That may be single stranded (ss) or **double** stranded (ds), **linear** or **circular**, **intact** or **segmented**.
- ***Functions of viral nucleic acid***
 - a. It is the essential infectious component of the virus.
 - b. It carries the heritable characters of the virus.
 - c. It codes for the production of new viral proteins and genomes.



2. Viral capsid:

Protein coat which surround **viral nucleic acid**.

- Made up of subunits called **capsomers**.
- The capsid with its enclosed nucleic acid is called **nucleocapsid**.

Function of viral capsid

1. It protects the viral nucleic acid.
2. Attachment of virus with susceptible host cell.
3. Responsible for the shape (symmetry) of the viruses.
4. Viral proteins are antigenic and immunogenic.

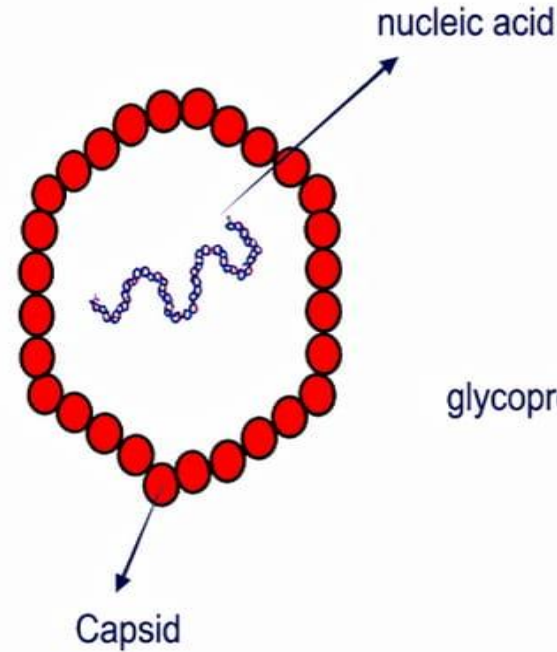


3. Viral envelope:

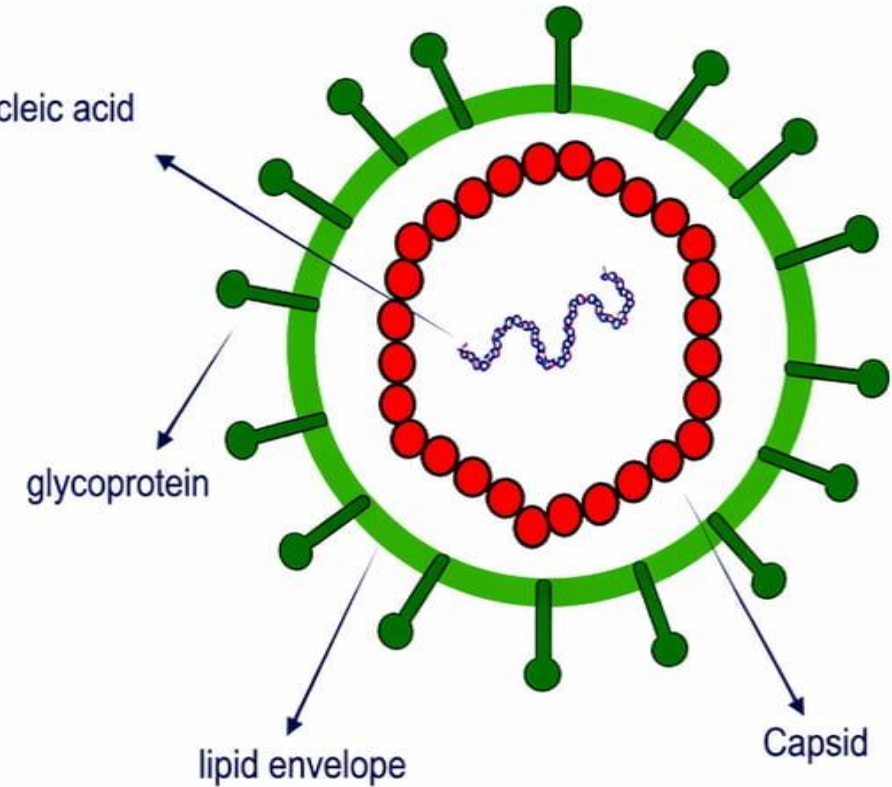
It is a **lipoprotein coat** acquired by some viruses as they **bud** from the **host cell** during their replication.

According to its presence or absence viruses can be classified into **enveloped** and **non-enveloped viruses**.

Non-enveloped virus



Enveloped virus



MCQ QUESTIONS

Q1.

Viruses are best described as:

- A) Independent living cells
- B) Multicellular organisms
- C) Obligate intracellular parasites
- D) Free-living bacteria

Q2.

Viral genetic material consists of:

- A) DNA only
- B) RNA only
- C) DNA and RNA together
- D) Either DNA or RNA

MCQ QUESTIONS

Q3.

Why can viruses not reproduce outside a host cell?

- A) They lack genetic material
- B) They lack host receptors
- C) They depend on host cell machinery
- D) They are too large

Q4.

The capsid of a virus mainly functions to:

- A) Produce energy
- B) Protect viral nucleic acid
- C) Destroy host cells
- D) Synthesize

MCQ QUESTIONS

Q5.

A nurse is handling a virus that is easily destroyed by detergents and drying. This virus is most likely:

- A) Non-enveloped virus
- B) Enveloped virus
- C) Bacterial cell
- D) Spore-forming organism

Q6.

A virus attaches to a host cell using surface proteins. Which structure is responsible?

- A) Nucleic acid
- B) Capsid only
- C) Glycoprotein spikes
- D) Ribosomes

Classification of Viruses

Viruses can be classified according to:

1

Type of the nucleic acid

- DNA viruses
- RNA viruses

2

Viral size

- Large viruses
- Medium viruses
- Small viruses

3

Viral shapes

- Helical
- Icosahedral
- Complex symmetry

4

Presence or absence of envelope

- Enveloped viruses
- Non-enveloped viruses

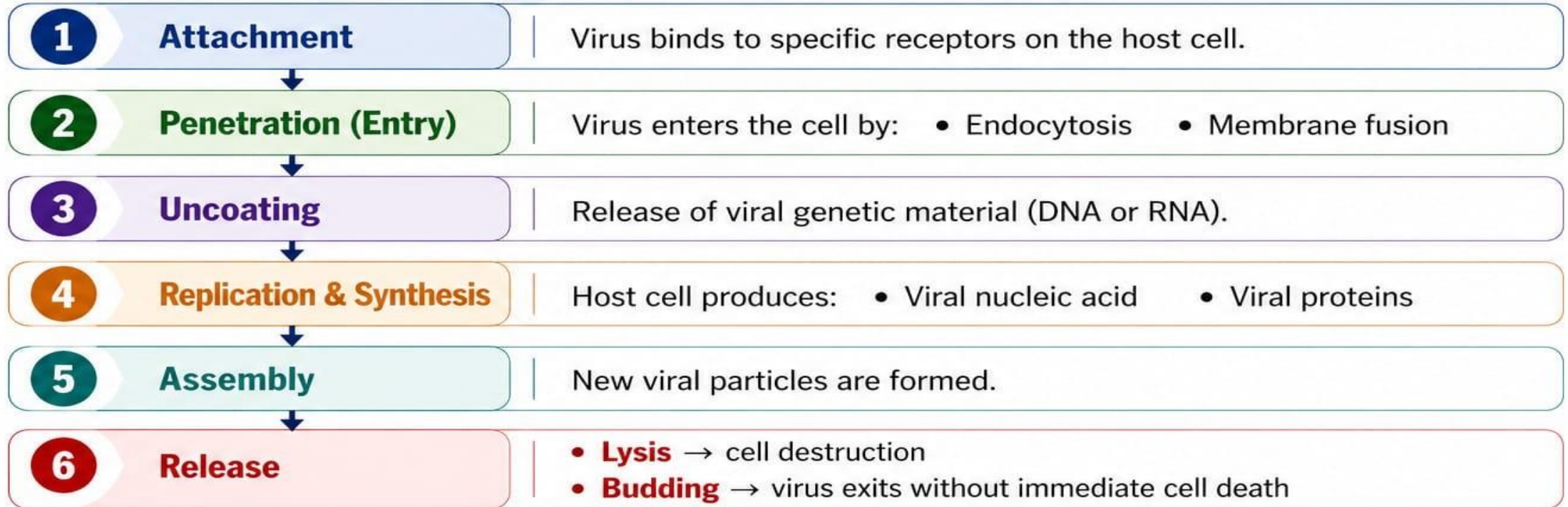
Viral Replication (Life Cycle)



Key Concept

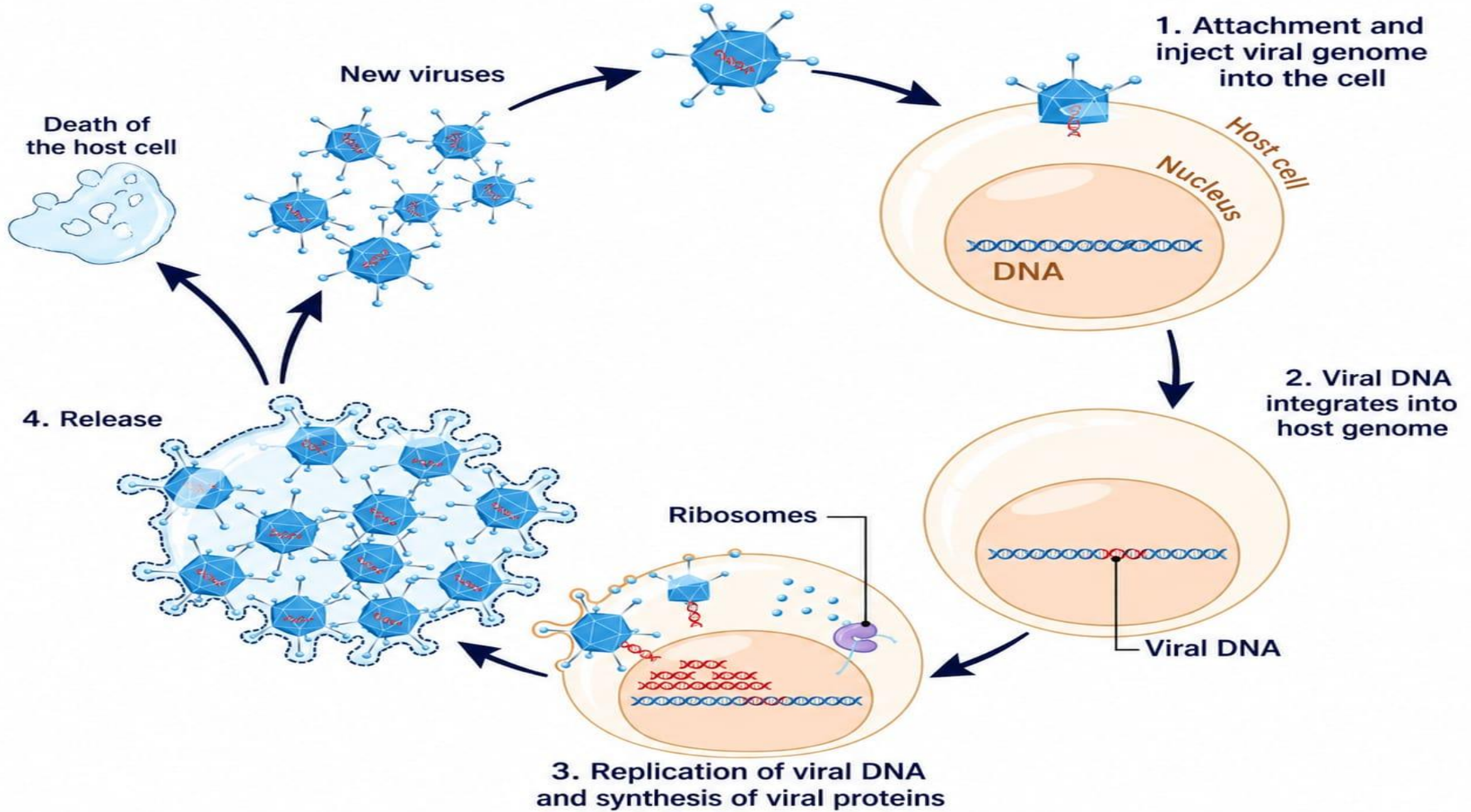
- Viruses do not divide like cells
- They replicate by **using host cell machinery**

Steps of Replication



Clinical Note: Viral replication occurs only inside host cells.

LIFE CYCLE OF VIRUSES



Pathogenesis of Viral Infections



HOW VIRUSES CAUSE DISEASE:



1 Cell destruction

Viruses can directly damage or kill host cells, leading to tissue injury and organ dysfunction.



2 Immune response damage

Excessive immune response to viruses can cause inflammation and tissue damage.



3 Latent infections

Some viruses remain inactive (latent) in the body and may reactivate later, causing disease.

TRANSMISSION ROUTES:



Respiratory
(e.g., influenza)



Fecal-oral
(e.g., hepatitis A)

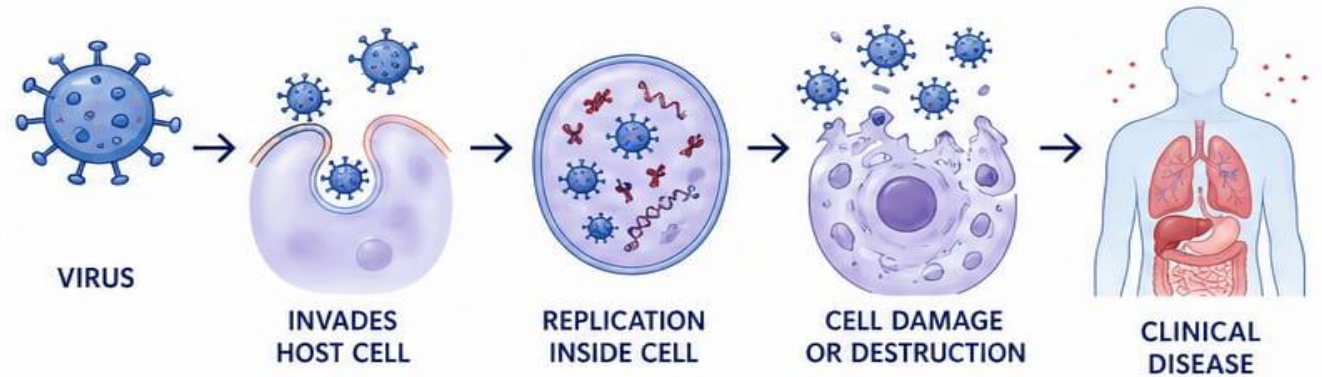


Blood
(e.g., HIV)



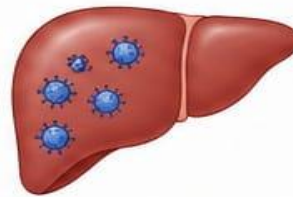
Direct contact
(e.g., herpes)

HOW VIRUSES LEAD TO DISEASE



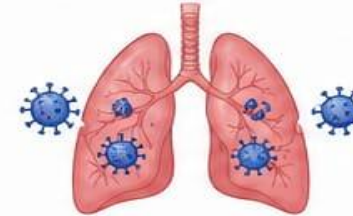
EXAMPLES OF VIRAL PATHOGENESIS

HEPATITIS B VIRUS



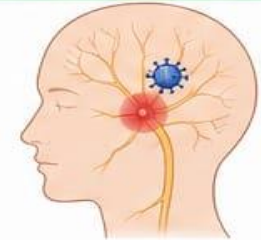
Infects liver cells → inflammation
→ chronic infection → cirrhosis
or liver cancer.

INFLUENZA VIRUS



Infects respiratory epithelium
→ cell destruction → fever,
cough, body aches.

HERPES SIMPLEX VIRUS



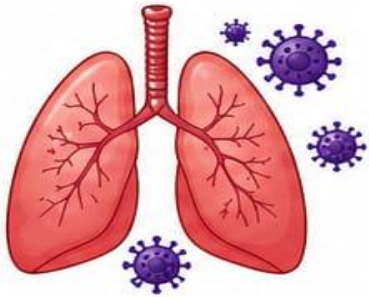
Establishes latency in nerve
cells → reactivation → cold
sores or genital lesions.

Viral Infections



Understanding key viral infections helps in prevention and control.

1 Influenza (Flu)



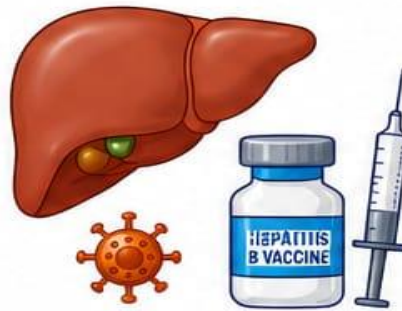
- Acute respiratory infection
- **Complications:** pneumonia
- **Prevention:** vaccine, hygiene

2 COVID-19



- Mild to severe disease
- **Complications:** ARDS, blood clots
- **Prevention:** mask, vaccination

3 Hepatitis B



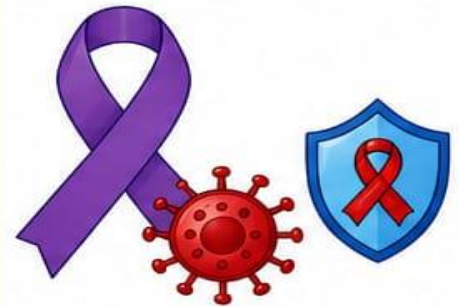
- Chronic liver disease
- **Complications:** cirrhosis
- **Prevention:** vaccine

4 Measles



- Highly contagious
- **Complications:** encephalitis
- **Prevention:** MMR vaccine

5 HIV/AIDS



- Attacks immune system
- Leads to opportunistic infections
- **Prevention:** safe practices



CLINICAL PEARL

Early detection + infection control + vaccination = **prevention of spread**

Common Viral Diseases



Viruses cause many important diseases. Nurses play a key role in prevention, early detection, and patient care.

Disease	Causative Virus	Transmissims	Main Symptoms	Key Nursing Actions
 Influenza	Influenza virus	Respiratory droplets	• Fever • Cough • Body aches	• Droplet precautions • Encourage fluids • Monitor complications
 COVID-19	SARS-CoV-2	Droplets, contact, aerosols	• Fever • Cough • Loss of smell • Shortness of breath • Fatigue	• Isolation precautions • Use PPE • Monitor oxygen saturation
 Hepatitis B	Hepatitis B virus (HBV)	Blood, body fluids, sexual contact	• Fatigue • Nausea • Jaundice • Abdominal pain	• Standard precautions • Safe injection practices • Promote vaccination
 Measles	Measles virus	Airborne (very contagious)	• Fever • Cough, runny nose • Red eyes • Rash	• Airborne precautions • Isolation • Vitamin A as ordered • Promote vaccination
 HIV/AIDS	Human Immunodeficiency Virus (HIV)	Blood, sexual contact, mother to child	• Fever • Weight loss • Night sweats • Recurrent infections	• Standard precautions • Patient education • Emotional support • Prevent stigma

IMPORTANT NURSING NOTES



Hand hygiene prevents spread of viruses.



Use appropriate PPE according to transmission route.



Follow isolation precautions to protect patients and staff.



Vaccination is the most effective way to prevent many viral diseases.



Polio Virus



GENERAL CHARACTERS

- Small non-enveloped RNA virus
- Icosahedral (cubic) capsid
- 3 antigenic types (1, 2, 3)



PATHOGENESIS

- Entry through the mouth
- Multiplication in **intestine/oropharynx**
- Spread via blood → **viremia**
- Reaches **CNS** → destroys **motor neurons**



CLINICAL MANIFESTATIONS

- **Flaccid paralysis**
- Muscle weakness (especially limbs)
- May affect **respiratory muscles**



KEY CONCEPT

Destruction of motor neurons →
loss of muscle function



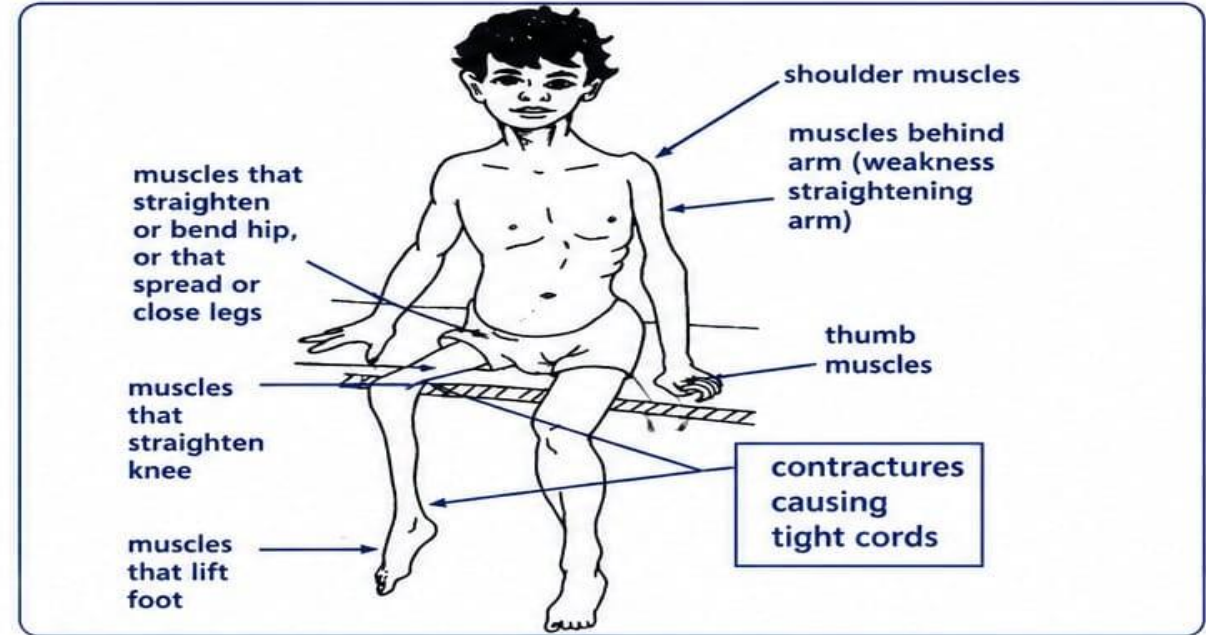
IMMUNITY

Infection leads to **long-lasting immunity** (type-specific).



CLINICAL PEARL

Poliovirus destroys motor neurons in the spinal cord → **flaccid paralysis**.





Mumps Virus



GENERAL CHARACTERS

- Enveloped, single-stranded **RNA** virus
- Large in size with **helical** symmetry
- One serological **type**



TRANSMISSION & PATHOGENESIS

- Spread by **droplet inhalation** or **oral route**
- Causes:
 - Fever
 - Swelling of one or both **parotid** and **submaxillary glands**



COMPLICATIONS

- Meningoencephalitis
- Orchitis / Oophoritis
- May lead to **infertility**



PREVENTION

- Mumps live attenuated vaccine available in combination with measles and rubella (**MMR**)
- Given **S.C.** (subcutaneous)
- One single dose after **9 months** up to **-15 months** of age



KEY CONCEPT

Viral infection of salivary glands → **Parotid swelling**



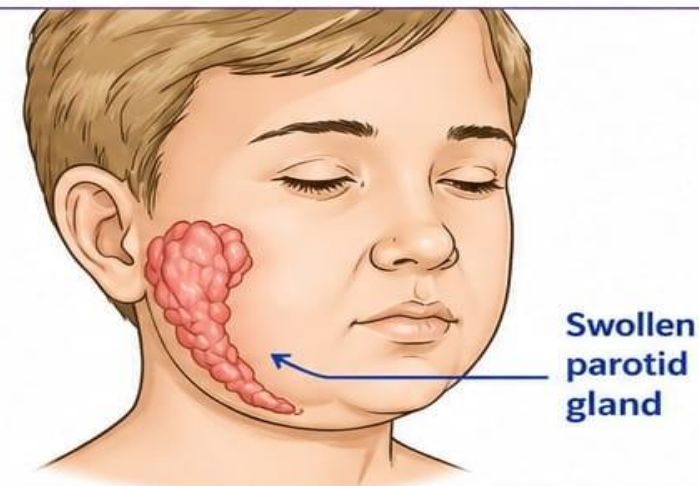
Mumps

is an infection caused by a virus that's easily spread through **saliva** and **mucus**.

It usually happens in children who haven't been immunized.

Mumps Affect Parotid Gland

The mumps virus infects the salivary glands, especially the **parotid gland** (in front of the ear), causing pain and swelling.



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Measles Virus



GENERAL CHARACTERS

- Large, single-stranded (SS) RNA virus
- Spherical shape with helical capsid symmetry
- One serological type



PATHOGENESIS

- Infection occurs by airborne inhalation
- Characterized by:
 - High fever
 - Skin rash
- May be complicated with otitis media, pneumonia or encephalitis
- One attack of measles is followed by life-long immunity.



PROPHYLAXIS

- MMR vaccine (live attenuated)
- Given S.C. in one single dose
- After 9 months up to 15 months of age



Measles

Measles is a highly contagious viral disease spread by airborne droplets.



One attack leads to life-long immunity.



KEY POINT

Airborne infection → High fever + Skin rash → May lead to complications
Prevention by MMR vaccine



Rubella Virus

(German Measles)



GENERAL CHARACTERISTICS

- Single-stranded RNA (**SS RNA**), enveloped
- Medium size with **helical symmetry**
- One serological **type**



PATHOGENESIS

- Transmitted by **respiratory droplets**
- Causes:
 - Mild **fever**
 - Maculopapular **rash** (starts on face → spreads)
- Infection during **early pregnancy** (first 16 weeks) leads to:
 - **Congenital Rubella Syndrome (CRS)**



CONGENITAL DEFECTS (CRS)

- **Deafness**
- **Cardiac abnormalities**
- **Cataract**
- **Mental retardation**



PREVENTION

- **MMR vaccine** (live attenuated)
- Provides **long-lasting immunity**



CLINICAL NOTE

Rubella is usually mild in children but **highly dangerous during pregnancy**

GERMAN MEASLES (rubella)



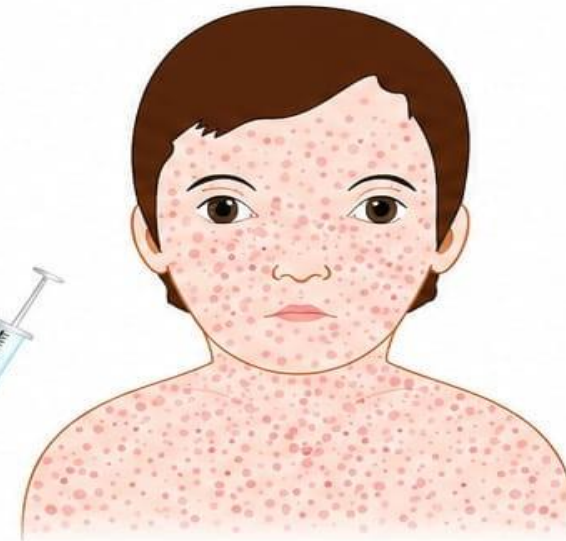
Rubivirus rubellae



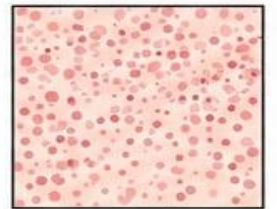
High fever



Vaccination



The rash causes itching and often lasts for about three days



Headache



Cold-like symptoms



Conjunctivitis



Joint pains

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MCQ QUESTIONS

Q1.

Which virus is associated with swelling of the parotid gland?

- A) Measles virus
- B) Rubella virus
- C) Mumps virus
- D) Polio virus

Q2.

Poliovirus primarily affects which part of the body?

- A) Liver cells
- B) Skin
- C) Kidney
- D) Motor neurons in CNS



MCQ QUESTIONS

Q3.

Why is measles considered highly contagious?

- A) It spreads through blood
- B) It spreads through airborne droplets
- C) It requires direct contact only
- D) It is transmitted through food

Q4.

Why is rubella infection dangerous during early pregnancy?

- A) Causes severe maternal fever
- B) Leads to congenital defects in fetus
- C) Causes respiratory failure
- D) Leads to kidney damage



MCQ QUESTIONS

Q5.

A patient develops flaccid paralysis after a viral infection. Which mechanism best explains this condition?

- A) Liver cell destruction**
- B) Skin inflammation**
- C) Destruction of motor neurons**
- D) Immune suppression**

Q6.

A pregnant woman in her first trimester is exposed to a viral infection. The fetus develops deafness and cardiac defects. Which virus is responsible?

- A) Measles virus**
- B) Rubella virus**
- C) Mumps virus**
- D) Poliovirus**



MCQ QUESTIONS

Q7.

Which preventive measure is **MOST** effective to protect against measles, mumps, and rubella?

- A) Antibiotic therapy
- B) Isolation only
- C) MMR vaccination
- D) Hand washing only

Q8.

A nurse is planning infection control measures for a patient with measles. Which is the **BEST** approach?

- A) Contact precautions only
- B) Airborne precautions
- C) No precautions needed
- D) Blood precautions only

